

COMPACT CY₃ FRONTIER: THEOREMS AND CONJECTURES OVER THE FORMAL-LOCAL MIXED HT SFT

I. MNOP FACTORISATION AND THE E₂-CENTRE OBSTRUCTION

On a smooth compact Calabi–Yau threefold X , three enumerative theories produce a priori distinct generating series: Donaldson–Thomas counts of ideal sheaves $Z_{\text{DT}}(q) = \sum_n \text{DT}_n(X) q^n$ (Thomas 2000; Behrend–Fantechi 2008), Pandharipande–Thomas stable-pair counts $Z_{\text{PT}}(q) = \sum_n \text{PT}_n(X) q^n$ (Pandharipande–Thomas 2009), and Gromov–Witten virtual-class series $Z_{\text{GW}}(u) = \sum_g \text{GW}_g(X) u^{2g-2}$ (Behrend–Fantechi 1997; Li–Tian 1998). The Maulik–Nekrasov–Okounkov–Pandharipande package asserts that the three series produce the same complex numbers under $-q = e^{iu}$ and a Katz–Klemm–Vafa integrality wedge; the proven half is the DT/PT factorisation by the MacMahon power off the degree-zero locus.

Theorem 1.1 (DT/PT factorisation on the quintic). *Let $Q \subset \mathbb{P}^4$ be a smooth quintic threefold (Euler characteristic $\chi(Q) = -200$, Calabi–Yau by $c_1(Q) = 0$). The degree-zero DT generating function factors as the MacMahon power $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ raised to $\chi(Q)$, and the reduced DT series equals the PT series:*

$$Z_{\text{DT}}(Q; q) = M(-q)^{\chi(Q)} Z_{\text{PT}}(Q; q).$$

Proof. The factorisation $Z_{\text{DT}} = Z_{\text{DT},0} \cdot Z'_{\text{DT}}$ with $Z_{\text{DT},0} = M(-q)^{\chi(Q)}$ is Behrend–Fantechi’s Hilbert-scheme formula applied to the degree-zero locus. The reduced equality $Z'_{\text{DT}} = Z_{\text{PT}}$ is Bridgeland’s Hall-algebra theorem, with Toda’s wall-crossing route as auxiliary. Smoothness of Q and $\chi(Q) = -200$ close the arithmetic. $\square$

The PT/GW leg $Z_{\text{PT}}(Q; q) = Z_{\text{GW}}(Q; u)|_{-q=e^{iu}}$ is conditional on KKV integrality for Q ; finite-genus evidence on the quintic requires the Huang–Klemm–Quackenbush bootstrap.

The substitution $-q = e^{iu}$ is not a tautology between two notations for the same generating function. At chain level, the DT, PT, and GW sides come from three a priori distinct dualisable modules over a factorisation-algebra image of $\text{Perf}(Q)$, each carrying its own partition function; a single chain-level automorphism must respect all three trace maps to produce the substitution. The compact target datum that carries this object is

$$C_{\text{tar}}(X) = (\mathcal{F}_X, M_{\text{DT}}, M_{\text{PT}}, M_{\text{GW}}, \text{Tr}_{\text{DT}}, \text{Tr}_{\text{PT}}, \text{Tr}_{\text{GW}}, \sigma_X),$$

where $\mathcal{F}_X = \Phi_3^{\text{FA}}(\text{Perf}(X))$ is the image of the 3-Calabi–Yau dg category of perfect complexes on X under a factorisation-algebra functor of degree 3; each $M_\bullet$ is a dualisable $\mathcal{F}_X$ -module attached to the corresponding theory; each $\text{Tr}_\bullet$ is a chain-level trace map recovering the scalar generating function; and $\sigma_X \in \text{Aut}_{E_2}(Z_{E_2}(\mathcal{F}_X))$ is an automorphism of the E₂-centre $Z_{E_2}(\mathcal{F}_X)$ of $\mathcal{F}_X$ (the centre under the Lurie E_n-centre formalism with $n = 2$, controlling braided-monoidal automorphisms) implementing the MNOP/PT–GW substitution.

Conjecture 1.2 (E_2 -centre MNOP package on the quintic). *Let $C_{\text{tar}}(Q)$ be constructed in conventions matched to the scalar fixture of Theorem 1.1. Then a unique $\sigma_Q \in \text{Aut}_{E_2}(Z_{E_2}(\mathcal{F}_Q))$ effects the substitution $-q = e^{iu}$ on $(\text{Tr}_{\text{DT}}, \text{Tr}_{\text{PT}}, \text{Tr}_{\text{GW}})$, and rigidity holds: any two such automorphisms agree in the E_2 -centre cohomology of $\mathcal{F}_Q$.*

The hypothesis decomposes into four constructions: the factorisation-algebra functor Φ_3^{FA} on a 3CY dg category (Joyce–Gross–Joyce–Tanaka Hall–vertex source vocabulary; Lurie centre formalism); the dualisable module structure on $M_{\text{DT}}, M_{\text{PT}}, M_{\text{GW}}$; the chain-level trace maps making each $Z_\bullet$ scalar; and a centrality homotopy $\sigma_X : \text{Tr}_{\text{DT}} \Rightarrow \text{Tr}_{\text{PT}}$ (and analogously to Tr_{GW}) in the E_2 -centre. Rigidity is the vanishing of an obstruction class $\text{Ob}_{\text{UKD}}(C_{\text{tar}})$ in the unique-Kuznetsov-descent group of the centre, controlling E_2 -uniqueness of σ_X under the dg/derived descent. Orientation and CoHA prerequisites enter through Joyce–Upmeyer and Kinjo–Park–Safronov.

2. S^3 -FRAMING ON THE QUINTIC

Costello’s chain-level open topological B-model with boundary on a Lagrangian $L \subset X$ requires a framing datum on L : a chosen trivialisation of the $SO(\text{rank } N_L)$ -torsor of orthonormal frames of the normal bundle, compatible with the brane-line BV complex. For a Lagrangian $L \cong S^3 \hookrightarrow Q$ in the smooth quintic -- for instance the real locus of a real deformation (Sheridan 2016; Sheridan–Smith 2021) -- the topological framing obstruction lives in

$$\pi_0(\text{Map}(S^3, SO(3))) = \pi_3(SO(3)) = \mathbb{Z},$$

the Hirzebruch σ -invariant torsor. Two questions separate: whether the Calabi–Yau structure on Q orients $N_{L/Q}$ and trivialises it topologically, so that the framing torsor is in fact $\mathbb{Z}$; and which integer in that torsor, paired with which BV/A_∞ chain homotopy on the brane complex, the open B-model requires.

Proposition 2.1 (Normal-bundle orientation and topological triviality). *Let $L \cong S^3 \subset Q$ be a Lagrangian in the smooth quintic threefold (Calabi–Yau by $K_Q = \mathcal{O}_Q$, with nowhere-vanishing holomorphic volume form $\Omega_Q \in H^0(Q, K_Q)$). Then $N_{L/Q}$ is canonically oriented and the oriented rank-three real bundle $N_{L/Q} \rightarrow S^3$ is topologically trivial. The orientation comes from*

$$\det T_L \otimes \det N_{L/Q} = \det(TQ|_L) \cong \mathcal{O}_L$$

via restriction of Ω_Q .

Proof. Since $K_Q = \mathcal{O}_Q$ and Ω_Q is nowhere zero, its restriction is a nowhere-vanishing section of $\det(TQ|_L)$. The Lagrangian splitting $TQ|_L = TL \oplus N_{L/Q}$ as real bundles gives $\det(TQ|_L)_{\mathbb{R}} = \det(TL) \otimes \det(N_{L/Q})$; parallelisability of S^3 makes $\det(TL)$ canonically trivial, so $\det(N_{L/Q})$ is trivial and $N_{L/Q}$ is oriented. Every oriented rank-three real bundle on S^3 is trivial: the classifying map lies in $BSO(3)$ and $\pi_3(BSO(3)) = \pi_2(SO(3)) = 0$. $\square$

The Calabi–Yau volume form supplies the topological orientation and triviality input. Two further data complete the Costello open B-model on (Q, L) : an integer framing class $[f] \in \pi_3(SO(3)) = \mathbb{Z}$ selecting a point in the trivialisation torsor, and a Costello-compatible BV/A_∞ chain homotopy h_f trivialising the obstruction class

$$\text{Ob}_{\text{frame}}(L, f) \in H^0(\Omega^\bullet(L) \otimes \text{End}_{\text{BV}})$$

to the equivariant lift of the brane propagator over the chosen framing. Framing on S^3 is a choice, not canonical: each $[f]$ defines a different chain-level brane.

3. CHIRAL-VOLUME FINITE EVIDENCE ON THE QUINTIC

If the genus-zero BPS counts n_N of degree- N rational curves on a compact Calabi–Yau threefold grow at the rate predicted by the period of a vanishing cycle, the sequence $a_N = N^{-1} \log n_N$ converges. The chiral-volume conjecture (calabi-yau-quantum-groups, Vol III) identifies the limit with the Hodge-normalised Abel–Jacobi period of a representative curve:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log |Z_N(X, \beta)| \stackrel{?}{=} \frac{1}{2\pi} |\mathbf{AJ}(C_\beta)|,$$

where $Z_N = Z_N(X, \beta)$ is the degree- N rational-curve count in class $\beta \in H_2(X, \mathbb{Z})$, $\mathbf{AJ}(C_\beta) \in J^2(X)$ is the Abel–Jacobi period of a representative curve C_β , and the intermediate Jacobian $J^2(X) = H^3(X, \mathbb{C}) / (F^2 H^3 + H^3(X, \mathbb{Z}))$ carries the Hodge metric. Take $X = Q$ the smooth quintic, $\beta = NH$ for $H \in H_2(Q, \mathbb{Z})$ the hyperplane class, $n_N = Z_N(Q, NH)$ the genus-zero BPS invariant. The finite data through $N = 10$ (BCOV bootstrap for $N \leq 9$, Huang–Klemm–Quackenbush 2009 for $N = 10$) reads:

$$\begin{aligned} n_1 &= 2875 \\ n_2 &= 609\,250 \\ n_3 &= 317\,206\,375 \\ n_4 &= 242\,467\,530\,000 \\ n_5 &= 229\,305\,888\,887\,625 \\ n_6 &= 248\,249\,742\,118\,022\,000 \\ n_7 &= 295\,091\,050\,570\,845\,659\,250 \\ n_8 &= 375\,632\,160\,937\,476\,603\,550\,000 \\ n_9 &= 503\,840\,510\,416\,985\,243\,645\,106\,250 \\ n_{10} &= 704\,288\,164\,978\,454\,686\,113\,488\,249\,750. \end{aligned}$$

The sequence $a_N := N^{-1} \log n_N$ attains its minimum at $N = 3$:

$$\begin{aligned} a_1 &\approx 7.964, \quad a_2 \approx 6.660, \quad a_3 \approx 6.525, \quad a_4 \approx 6.554, \quad a_5 \approx 6.613, \\ a_6 &\approx 6.676, \quad a_7 \approx 6.733, \quad a_8 \approx 6.785, \quad a_9 \approx 6.832, \quad a_{10} \approx 6.873. \end{aligned}$$

The sequence decreases sharply through $N = 3$, then increases monotonically: $a_{N+1} - a_N > 0$ for $N \in \{3, \dots, 9\}$, with increments

$$\{+0.029, +0.060, +0.062, +0.058, +0.052, +0.046, +0.041\}.$$

Proposition 3.1 (Quintic chiral-volume finite evidence through $N = 10$). *Take the displayed genus-zero quintic BPS table through $N = 10$ as input. The sequence $a_N = N^{-1} \log n_N$ attains its minimum at $N = 3$ and, on the tail $N \in \{5, \dots, 10\}$, satisfies the three-parameter law*

$$a_N = L + \alpha \frac{\log N}{N} + \frac{\beta}{N}$$

with residual bounded by 5×10^{-6} and

$$L \approx 7.5925, \quad \alpha \approx -3.3204, \quad \beta \approx 0.4477.$$

The intercept L agrees, to three significant figures, with the raw mirror-period ratio at the conifold point $z_c = 5^{-5}$ of the Candelas–de la Ossa–Green–Parkes (CDGP) mirror quintic:

$$|t(z_c)| = \left| \frac{\omega_1(z_c)}{\omega_0(z_c)} \right| \approx 7.5909,$$

where the fundamental period $\omega_0(z) = \sum_n \binom{5n}{n,n,n,n,n} z^n$ and the logarithmic period $\omega_1(z) = \omega_0(z) \log z + 5 \sum_n \binom{5n}{n,n,n,n,n} (H_{5n} - H_n) z^n$ (with H_k the k -th harmonic number) solve the quintic Picard–Fuchs operator $\mathcal{L}_{\text{PF}}$ of (1) below.

Proof. Linear least squares on the six equations $a_N = L + \alpha(\log N)/N + \beta/N$, $N \in \{5, \dots, 10\}$, gives the stated coefficients with maximum residual $4.62 \times 10^{-6} < 5 \times 10^{-6}$. The fundamental and logarithmic periods solve

$$(1) \quad \mathcal{L}_{\text{PF}} = \theta^4 - 5z(5\theta + 1)(5\theta + 2)(5\theta + 3)(5\theta + 4), \quad \theta = z\partial_z,$$

with $z = 5^{-5}\psi^{-5}$ the natural mirror coordinate. At $z_c = 5^{-5}$ the convergent power-series sum (verified against scripts/quintic_arithmetic_rate_checks.py) yields

$$\omega_0(z_c) \approx 1.0707258684, \quad \omega_1(z_c) \approx -8.1277650532,$$

giving $|t(z_c)| \approx 7.5908925831$. □

Three numerical values stand together: the tail-fit intercept $L \approx 7.59$, the raw mirror-period ratio $|t(z_c)| \approx 7.5909$, and the OSV attractor comparison value $5 \log 5 \approx 8.05$. The fitted correction $\alpha \approx -3.32 < 0$ forces approach to L from below, so the conjectural chiral-volume limit is not bounded above by $\{a_1, a_2, a_3\}$; those low-degree points sit on the Schubert–Katz–Ellingsrud–Strømme Fano-geometric branch, below the large- N BPS regime.

The chiral-volume limit $\lim a_N = (2\pi)^{-1} |\mathbf{AJ}(C_H)|$ is stated against the Hodge metric on $J^2(Q)$. The raw CDGP mirror ratio $t(z_c) = \omega_1(z_c)/\omega_0(z_c)$ is not an Abel–Jacobi norm: ω_0, ω_1 sit in a CDGP basis of solutions of (1) that has not been Hodge-normalised, and the $2\pi i$ monodromy factor is absent. Bridging the two requires a normalisation datum

$$N_{\mathbf{AJ}}: (\omega_0, \omega_1, \omega_2, \omega_3) \mapsto (\varpi_0, \varpi_1, \varpi_2, \varpi_3)$$

matching the CDGP period basis to the Hodge filtration on H^3 and fixing the $2\pi i$ monodromy convention. Without $N_{\mathbf{AJ}}$, the finite agreement $L \approx |t(z_c)|$ is a normalisation check, not a proof of the limit.

4. SERRE PAIRING AS COMPACT-CY CATEGORICAL INPUT

The Serre functor and its pairing are the categorical Calabi–Yau datum that any compact CY₃ descent must preserve, before any module, tilting, or Hall-algebra structure is built. Van den Bergh’s noncommutative crepant resolution (NCCR) is defined for singular Gorenstein CY₃ via an Azumaya-like generator of D^b ; it is Iyama–Wemyss in the toric case and a consequence of the Orlov blow-up equivalence at $K_3 \times E$. On the smooth quintic there is no crepant resolution to resolve, and the NCCR question degenerates to the bare Serre datum.

Proposition 4.1 (Serre pairing on smooth compact CY₃). *Let X be a smooth compact Calabi–Yau threefold. The Serre functor $S_X: D^b(X) \rightarrow D^b(X)$ is the shift $S_X = [3]$, and the Calabi–Yau pairing*

$$\mathrm{RHom}(F, G) \otimes \mathrm{RHom}(G, F) \longrightarrow \mathbb{C}[-3]$$

is non-degenerate on $\mathrm{Perf}(X) = D^b(X)$.

Proof. Smoothness gives $\mathrm{Perf}(X) = D^b(X)$. Serre duality is $S_X = \otimes K_X[\dim X]$; the Calabi–Yau condition $K_X = \mathcal{O}_X$ reduces this to $S_X = [3]$. The pairing is the Serre trace of $\mathrm{RHom} \otimes \mathrm{RHom} \rightarrow \mathrm{RHom}(F, F)$ followed by the Serre trace $\mathrm{RHom}(F, F) \rightarrow \mathbb{C}[-3]$. $\square$

The Serre datum $S_X = [3]$ is what a compact-CY descent must preserve. The remaining global quasi-NCCR obstruction is the construction of a Serre-equivariant local algebraic or dg model $\mathcal{A}_X \rightarrow \mathrm{End}_{D^b(X)}(\bigoplus_i F_i)$ generating $D^b(X)$, with finite-dimensionality of $\mathrm{Ext}^*(F_i, F_j)$ and a descent datum identifying the local model with a global endomorphism sheaf compatibly with S_X . Such an object exists in the singular Gorenstein cases above; on the smooth quintic it is open.

5. FOURIER–MUKAI PARTNERS FOR $K_3 \times E$

Kuznetsov homological projective duality (Kuznetsov 2007) takes as input a smooth projective variety with a very ample line bundle inducing a non-trivial Lefschetz decomposition of D^b ; on any Calabi–Yau the Serre functor is a shift, and Bridgeland’s indecomposability theorem (Bridgeland 1999) constrains the only semiorthogonal decomposition to the trivial one. Both $D^b(K_3)$ and $D^b(E)$ are indecomposable, so the Lefschetz input on $K_3 \times E$ is the trivial rectangular structure $\mathcal{A}_0 = D^b(K_3 \times E)$, and Kuznetsov Fano HPD does not apply. The derived-equivalence statement on $K_3 \times E$ takes a different shape: Fourier–Mukai partner construction governed by Mukai-vector coprimality.

Theorem 5.1 (Fourier–Mukai equivalence in the isotropic $K_3 \times E$ case). *Let S be a projective K_3 surface with primitive polarisation H_S , and let E be a principally polarised elliptic curve. Fix a Mukai vector $v = (r, \ell, s) \in \tilde{H}(S, \mathbb{Z})$ with $\ell \in \mathrm{NS}(S)$, Mukai pairing $\langle v, v \rangle = 0$, and H_S chosen v -generic. Assume the coprimality condition*

$$(\star) \quad \gcd(r, \ell \cdot H_S, s) = 1.$$

Let $\mathcal{M} := M_{H_S}(v)$ be the moduli space of H_S -stable sheaves with Mukai vector v . Then:

- (i) *$\mathcal{M}$ is a smooth projective K_3 surface (Mukai 1987; O’Grady 1997);*
- (ii) *there is an equivalence of $\mathbb{C}$ -linear triangulated categories*

$$\Phi_{\mathcal{U}} : D^b(S) \xrightarrow{\sim} D^b(\mathcal{M}),$$

given by Fourier–Mukai transform along the universal sheaf $\mathcal{U}$ on $S \times \mathcal{M}$ (Bridgeland–Maciocia 2001, Complex surfaces with equivalent derived categories, Thm. 1.1);

- (iii) *combined with the Poincaré–Mukai autoequivalence $\mathrm{FM}_{\mathcal{P}} : D^b(E) \xrightarrow{\sim} D^b(\hat{E}) = D^b(E)$ (Mukai 1981), the product transform*

$$\Phi_{\mathcal{U}} \boxtimes \mathrm{FM}_{\mathcal{P}} : D^b(S \times E) \xrightarrow{\sim} D^b(\mathcal{M} \times E)$$

is an equivalence of CY₃ categories preserving the Serre structure $S_X = [3]$. When $M \simeq S$ this is an untwisted Fourier–Mukai self-equivalence; in general it is a derived equivalence to the partner $M \times E$.

Proof. (1) Under $(\star)$ and v -genericity of H_S , every H_S -semistable sheaf with Mukai vector v is stable (no strictly semistables); the moduli M is then a smooth projective holomorphic symplectic variety of dimension $\langle v, v \rangle + 2 = 2$. Hence M is a K_3 surface. For positive Mukai square the moduli space is higher-dimensional hyperkähler and the product with E is no longer a CY₃ of the required form.

(2) The universal sheaf $\mathcal{U}$ exists on $S \times M$ under $(\star)$: $\gcd(r, \ell \cdot H_S, s) = 1$ kills the Brauer obstruction to a universal family (Căldăraru 2000, Thm. 1.3.2). The Fourier–Mukai kernel $\mathcal{U} \in D^b(S \times M)$ satisfies the Bridgeland–Maciocia orthogonality criterion (Bridgeland–Maciocia 2001, §1) and induces an equivalence. Without $(\star)$, $\mathcal{U}$ exists only as a $p_M^* \alpha$ -twisted sheaf for a Brauer class $\alpha \in \text{Br}(M)$, yielding instead $D^b(S) \simeq D^b(M, \alpha)$ (Huybrechts–Stellari 2005).

(3) On E with principal polarisation, $\hat{E} \cong E$ canonically; the Poincaré bundle $\mathcal{P} \in D^b(E \times E)$ is the kernel of $\text{FM}_{\mathcal{P}}$ (Mukai 1981, Thm. 2.2). Tensoring kernels, $\mathcal{K} := \mathcal{U} \boxtimes \mathcal{P} \in D^b((S \times E) \times (M \times E))$ defines a Fourier–Mukai transform whose composition with its adjoint is the identity on each factor; the external-tensor convolution $(\Phi_1 \boxtimes \Phi_2) \circ (\Psi_1 \boxtimes \Psi_2) = (\Phi_1 \circ \Psi_1) \boxtimes (\Phi_2 \circ \Psi_2)$ for FM kernels (Huybrechts, *Fourier–Mukai transforms in algebraic geometry*, Prop. 5.10) lifts the factor identities to the product. The Serre functor is preserved by any exact equivalence of triangulated categories (uniqueness of the Serre functor up to natural isomorphism); since both $\Phi_{\mathcal{U}}$ and $\text{FM}_{\mathcal{P}}$ are equivalences between CY categories of dimensions 2 and 1 respectively, the product Serre $S_{S \times E} = S_S \boxtimes S_E = [2] \boxtimes [1] = [3]$ survives. $\square$

Proposition 5.2 (Brauer-twisted dichotomy in the isotropic case). *Keep the isotropic hypothesis $\langle v, v \rangle = 0$ and drop $(\star)$ in Theorem 5.1. Then there exists a Brauer class $\alpha \in \text{Br}(M)$, whose order divides $\gcd(r, \ell \cdot H_S, s)$, such that*

$$D^b(S \times E) \simeq D^b((M, \alpha) \times E),$$

where the right-hand side denotes the derived category of $p_M^* \alpha$ -twisted sheaves. The partner is $(M, \alpha) \times E$ with twisted K_3 factor; an untwisted kernel exists only under the coprimality hypothesis.

Proof. Căldăraru 2000 Thm. 1.3.2 constructs the Brauer obstruction to the universal sheaf from the non-coprime Mukai vector; its order is the gcd. Huybrechts–Stellari 2005 (*Equivalences of twisted K_3 surfaces*) then identifies the derived equivalence as a twisted Fourier–Mukai. The elliptic factor contributes an untwisted $\text{FM}_{\mathcal{P}}$. $\square$ $\square$

The Kuznetsov–Perry 2019 categorical-join / rectangular-HPD machinery does not upgrade Fourier–Mukai to Fano HPD here, precisely because neither $D^b(K_3)$ nor $D^b(E)$ admits a non-trivial Lefschetz decomposition to feed into the rectangular construction.

Remark 5.3 (Distinct κ values on $K_3 \times E$ and its anchors). Coprimality $(\star)$ is the primitive Mukai-vector hypothesis controlling the untwisted Fourier–Mukai kernel. Four κ values attached to $K_3 \times E$ and its comparison anchors are pairwise distinct:

- $\kappa_{\text{cat}}(K_3 \times E) = \chi(\mathcal{O}_{K_3 \times E}) = 0$, the total-space categorical Euler invariant by Künneth.
- $\kappa_{\text{fibre}}(K_3) = \chi(\mathcal{O}_{K_3}) = 2$, the fibre Euler datum.
- $\text{rk } \tilde{H}^*(K_3) = 24$, the rank of the extended Mukai lattice $\tilde{H}^*(K_3, \mathbb{Z}) = H^0 \oplus H^2 \oplus H^4$.
- $\kappa_{\text{ch}}^{\text{Heis}} = 3$ and $\kappa_{\text{BKM}}(\Phi_1) = 5$ in the paramodular Igusa Δ_5 row.

No equality among these values follows from Theorem 5.1.

6. NON-PERTURBATIVE BCOV: GOPAKUMAR–VAFA CLOSURE

The perturbative BCOV genus expansion on a compact Calabi–Yau threefold is a divergent formal series of Gevrey class 1: genus- g amplitudes grow as $(2g-3)!/A^{2g-2}$ with $A = \oint_{\gamma_c} \Omega$ the period of a vanishing cycle at the nearest conifold singularity (Shenker; Mariño et al.). Borel summation on the physical ray fails; a non-perturbative completion is required. Two candidates compete: the Gopakumar–Vafa BPS resummation, and a global resurgent Stokes sheaf.

Costello–Li 2012 (arXiv:1201.4501) construct the classical BCOV field theory and the perturbative BV quantisation problem, with elliptic-curve uniqueness and Virasoro results within the stated scope; no compact CY₃ all-genus theorem for the quintic amplitudes follows. Write $\{F_g^{\text{BCOV}}(X, t)\}_{g \geq 0}$ for a solution of the BCOV holomorphic anomaly equation with standard large-radius and conifold-gap boundary data; the perturbative generating function is

$$F^{\text{pert}}(\lambda, t) = \sum_{g \geq 0} \lambda^{2g-2} F_g(X, t), \quad \lambda = g_s.$$

The Gopakumar–Vafa schema (Gopakumar–Vafa 1998) proposes the non-perturbative form

$$(2) \quad F^{\text{GV}}(\lambda, t) = \sum_{\beta > 0} \sum_{g \geq 0} n_{\beta}^g \sum_{m \geq 1} \frac{1}{m} (2 \sin(m\lambda/2))^{2g-2} e^{-m\beta \cdot t},$$

where $n_{\beta}^g \in \mathbb{Z}$ are integer BPS invariants of the class β . Each fixed- β summand resums infinitely many perturbative genera and produces Schwinger poles at $\lambda = 2\pi k/m$ whose residues are pair-production amplitudes in the dual gauge description.

Conjecture 6.1 (GV non-perturbative closure on the quintic). *Let $Q \subset \mathbb{P}^4$ be the smooth quintic, $F_g^{\text{BCOV}}(Q, t)$ the genus- g solution of the BCOV holomorphic anomaly recursion with CDGP 1991 large-radius input and standard conifold gap boundary condition, and F^{GV} the Gopakumar–Vafa resummation (2) with BPS invariants $n_{\beta}^g(Q)$ from Huang–Klemm–Quackenbush 2009 through genus 22. In the Kähler cone $\Re t > t_c$, where t_c is the image of the conifold locus $\psi = 1$ on the complex-structure moduli space of the mirror quintic $\tilde{Q}$:*

(i) *For every $0 \leq g \leq 22$,*

$$F_g^{\text{BCOV}}(Q, t) = [\lambda^{2g-2}] F^{\text{GV}}(\lambda, t)$$

as formal power series in $e^{-\beta \cdot t}$; equality holds in $\mathbb{Z}[[e^{-t}]]$ after the Katz–Klemm–Vafa integrality normalisation.

(ii) *The genus-0 BPS series $\sum_{\beta} n_{\beta}^0 e^{-\beta \cdot t}$ has a nonempty convergence domain, controlled by the conifold/entropy scale.*

(iii) The trans-series completion of F^{pert} produced by the GV resummation is compatible near the conifold point $\psi = 1$ with the Candelas–de la Ossa–Green–Parkes 1991 mirror-period monodromy: the local Stokes normalisation is $S_c = 1$, and the instanton action is $A = 2\pi i \tau_c$, where τ_c is the vanishing-cycle period $\oint_{\gamma_c} \Omega_{\tilde{Q}}$.

Clause (i) applies the Pandharipande–Thomas leg of MNOP (Theorem 1.1) to the BCOV genus- g amplitudes determined by the holomorphic anomaly recursion and boundary data; this is not a compact-CY₃ Costello–Li all-genus theorem, and the CDGP prepotential, HKQ genus-22 bootstrap, GV sine expansion, and PT/GW matching enter as inputs. Clause (ii) is a convergence problem for the genus-zero BPS series: Castelnuovo-type genus bounds do not control the number n_β° of rational curves, so finite-genus and finite-degree data alone cannot bound the coefficient growth of $\sum_\beta n_\beta^\circ e^{-\beta t}$; the open analytic object is a coefficient-growth theorem. Clause (iii) is the local conifold Stokes match: the leading factorial growth $F_g^{\text{pert}}(Q, t) \sim (2g-3)!/(2\pi i \tau_c)^{2g-2}$ along the Borel direction fixed by the vanishing cycle γ_c (with $\tau_c = \oint_{\gamma_c} \Omega_{\tilde{Q}}$) identifies, by Strominger’s conifold mechanism, the residue at this Borel singularity with a single hypermultiplet of mass $|\tau_c|$; in (2) this is the leading- m contribution of a single BPS hypermultiplet with $n_{\beta_c}^\circ = 1$ and $\beta_c \in H_2(Q, \mathbb{Z})$ the A-model class paired with γ_c under mirror symmetry, scoped to a neighbourhood of the conifold ray, with the Écalé–Borel resurgent sum giving Stokes normalisation $S_c = 1$ across that ray.

Conjecture 6.2 (Conifold trinity on the quintic). *On the quintic $Q \subset \mathbb{P}^4$, three numerical data coincide as complex numbers up to sign and overall $2\pi i$ normalisation:*

$$A_{\text{BCOV}} = A_{\text{Schwinger}} = A_{\text{period}},$$

where A_{BCOV} is the Shenker–Mariño instanton action of the perturbative BCOV series, read off as $|F_g|/|F_{g+1}| \sim |A|^2/(2g-2)(2g-1)$; $A_{\text{Schwinger}} = 2\pi i/m$ is the Schwinger-pole location of the Gopakumar–Vafa summand at multiplicity m , rescaled by $\beta_c \cdot t_c$; and $A_{\text{period}} = \oint_{\gamma_c} \Omega_{\tilde{Q}}$ is the vanishing-cycle period at the conifold locus $\psi = 1$ on the mirror quintic. Equality of all three forces the conifold Stokes normalisation $S_c = 1$ and a single BPS hypermultiplet $n_{\beta_c}^\circ = 1$.

Equality $A_{\text{BCOV}} = A_{\text{period}}$ follows from a Huang–Klemm–Quackenbush/Shenker–Mariño large-order computation; equality $A_{\text{Schwinger}} = A_{\text{period}}$ follows from the Vafa/Schwinger pair-production identity and the Strominger conifold mechanism; the resurgent normalisation $S_c = 1$ follows from the Cecotti–Codesido–Mariño row.

Conjecture 6.1(iii) Borel-sums along the physical ray and matches at the single conifold Stokes wall. Full non-perturbative completion requires matching at every Stokes wall of the quintic moduli space: large complex structure (where (2) is the target definition), the conifold (where Strominger’s hypermultiplet pairs with the vanishing cycle), and Landau–Ginzburg / Gepner (where the HKQ bootstrap supplies finite-genus data). The carrier object is a global resurgent sheaf $\mathcal{R}^{\text{np}}$ on the moduli space whose unique trans-series $F^{\text{np}}(\lambda, t) \in \Gamma(\mathcal{M}, \mathcal{R}^{\text{np}})$ has Stokes jumps matching the BCOV monodromy representation on $H^3(\tilde{Q}, \mathbb{Z})$; construction of $\mathcal{R}^{\text{np}}$ is open, both globally on the quintic moduli space and in any compact CY₃ generalisation.

Conjecture 6.3 (Mirror-period recovery from non-perturbative BCOV). *Let $\varpi_0(z), \varpi_1(z), \varpi_2(z), \varpi_3(z)$ be the Hodge-normalised CDGP 1991 fundamental periods of the mirror quintic $\widetilde{Q}$, solutions of the Picard–Fuchs operator $\mathcal{L}_{\text{PF}}$ of (1), with the mirror coordinate $z = \varsigma^{-5}\psi^{-5}$ and conventions matched to those of Proposition 3.1. The genus-zero prepotential extracted from $F^{\text{GV}}(\lambda, t) = \lambda^{-2}\mathcal{F}_0(t) + O(\lambda^0)$ with mirror map $t = \varpi_1/\varpi_0$ satisfies*

$$\mathcal{F}_0(t) = \frac{1}{\varpi_0^2}(\varpi_0\varpi_3 - \varpi_1\varpi_2) + (\text{classical polynomial}),$$

with no instanton corrections beyond those encoded in $\{n_\beta^0\}_\beta$. The first three BPS values $n_1^0 = 2875$, $n_2^0 = 609\,250$, $n_3^0 = 317\,206\,375$ come from the BCOV table; the CDGP period/prepotential row and the Schubert–Katz–Ellingsrud–Strømme historical mirrors enter as inputs.

Under the Hodge-normalisation datum N_{AJ} , the Hodge-normalised Abel–Jacobi period coincides with the mirror-coordinate modulus $|t(z_c)|$, and the conjectural chiral-volume limit L of Proposition 3.1 equals the logarithmic period $|\varpi_1(z_c)/\varpi_0(z_c)|$ of the CDGP family at the conifold attractor. Finite evidence comes from the HKQ genus-22 data; higher-genus extension requires the resurgent completion of Conjecture 6.1(iii).